



QXU5010

Module Name: Surfaces and Interfaces

QMES Examinations 2024/25 Semester A

Time allowed 2 hours

Write ALL the answers in the Answer Book

Complete the information below about yourself very carefully.

QM student number

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NPU student number

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Class number

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NOT allowed: electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. **Write only with a black or blue pen and in English.**
3. **Do all rough work in the answer book – do not tear out any pages.**
4. **If you use Supplementary Answer Books, tie them to the end of this book.**
5. **Write clearly and legibly.**
6. **Read the instructions on the inside cover.**

Examiners

Dr Colin Crick and Dr Shafique Ahmed

Instructions

Before the start of the examination

- 1) Place your student identity card on the corner of your desk so that your picture is visible.
- 2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance. Possession of mobile phones, electronic devices and unauthorised materials is an offence.**
- 3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**
- 4) Do not turn over your question paper or begin writing until told to do so.

During the examination

- 1) You must not communicate with (by speaking or signs or by any other way) or copy from another student.
- 2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.
- 3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.
- 4) For 2-hour examinations you may **not** leave temporarily.
- 5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

- 1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**
- 2) Remain in your seat until you are told you may leave

Multiple Choice Questions. For each question, choose the answer that you think is correct and mark the answer sheet clearly with your choice. There is only 1 correct answer to each question. Each correct answer is worth 1 mark. [20 marks]

Q1. What is the primary reason surfaces behave differently from bulk materials?

- a) Surfaces have fewer atoms than bulk materials
- b) Surfaces exhibit different optical properties
- c) Surfaces have unbalanced forces at the boundary
- d) Surfaces have higher temperatures than the bulk

Q2. Which of the following is a characteristic of hydrophobic surfaces?

- a) Strong affinity for water
- b) Low contact angle with water
- c) High contact angle with water
- d) Dissolves easily in water

Q3. Which type of interaction is dominant in physisorption?

- a) Ionic bonds
- b) Covalent bonds
- c) Van der Waals forces
- d) Chemical reactions

Q4. What type of film does the Langmuir-Blodgett technique typically produce?

- a) Three-dimensional crystalline film
- b) Two-dimensional monolayer film
- c) Powdered surface film
- d) Absorbed film in bulk material

Q5. What happens to the surface area-to-volume ratio as particle size decreases?

- a) It decreases
- b) It remains constant
- c) It increases
- d) It fluctuates randomly

Q6. Which of the following is a key concept in the DLVO theory?

- a) Surface roughness
- b) Aggregation and flocculation
- c) Adsorption at solid-gas interface
- d) Surface energy minimization

Q7. Which type of adsorption typically involves higher energy and the formation of chemical bonds?

- a) Physisorption
- b) Chemisorption
- c) Van der Waals adsorption
- d) Ionic adsorption

Q8. Which of the following processes is NOT a method of surface cleaning?

- a) Ion bombardment
- b) Chemical treatment
- c) Adsorption of contaminants
- d) Heating to remove oxides

Q9. What happens at the Critical Micelle Concentration (CMC) of surfactants?

- a) Surfactants precipitate out of solution
- b) Surface tension remains constant despite increasing surfactant concentration
- c) Surfactants decompose
- d) Surfactants evaporate rapidly

Q10. Which of the following can result from surface defects such as steps, kinks, or vacancies?

- a) Reduced surface reactivity
- b) Increased chemical reactivity due to higher energy sites
- c) Uniform adsorption across the surface
- d) Decreased adsorption

Q11. Which of the following summarises the role of the electron beam (e-beam) in e-beam physical vapour deposition?

- a) The e-beam probes the elemental composition of the material
- b) The e-beam heats and vapourises the source material
- c) The e-beam quantifies the conductivity of the material
- d) The e-beam ensures an inert atmosphere for the deposition

Q12. Which of the following coating techniques/deposition mechanisms tends to generate rough surfaces on a flat substrate?

- a) Chemical Vapour Deposition (CVD) – layer-by-layer growth
- b) Atomic Layer Deposition (ALD)
- c) Langmuir-Blodgett (LB) deposition
- d) CVD – island growth

Q13. Which of the following factors does NOT generally affect if two liquids are miscible?

- a) The type of intermolecular attraction within each liquid phase
- b) The time the liquids are left to interact with each other
- c) The molar ratio of the two liquids
- d) The temperature of the two liquids

Q14. In the following examples, two liquids are to be mixed, which are most likely to be miscible?

- a) Water (H_2O) and Methanol (CH_3OH)
- b) Water (H_2O) and n-Hexane ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$)
- c) Water (H_2O) and Xylene ($(\text{CH}_3)_2\text{C}_6\text{H}_4$)
- d) n-Hexane ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$) and Methanol (CH_3OH)

Q15. If a flat surface is covered with a monolayer of the following chemical groups, which would be expected to have the highest water contact angle?

- a) $-\text{COOH}$
- b) $-\text{CH}_2\text{OH}$
- c) $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- d) $-\text{CH}_3$

Q16. Which of the following wetting models does NOT allow for a rough wetted surface NOR the presence of air?

- a) Young
- b) Wenzel
- c) Cassie-Baxter
- d) Simplified Cassie-Baxter

Q17. What is the main purpose of measuring the water contact angle hysteresis ($\theta_A - \theta_R$) of a surface?

- a) $\theta_A - \theta_R$ alone is the best measure of surface hydrophobicity
- b) $\theta_A - \theta_R$ provides an insight into the dynamic wetting behaviour
- c) $\theta_A - \theta_R$ alone confirms if the surface is superhydrophobic
- d) $\theta_A - \theta_R$ provides a quantitative measure of surface roughness

Q18. Which of the following approaches can generally be used to convert a hydrophobic surface into a superhydrophobic one?

- a) Make the surface chemistry more polar
- b) Make the surface flatter
- c) Make the surface rougher
- d) Change the wetting model used to examine the surface

Q19. Which of the following is principally considered when preventing the cohesive failure of an adhesive joint?

- a) The strength of adhesive-adherend chemical adhesion
- b) The type of material used for adherends
- c) The strength of adhesive-adherend mechanical adhesion
- d) The wetting of an adhesive on the adherends

Q20. Which of the following surface roughness measurements does NOT indicate surface roughness scale (i.e. nanoscale or microscale roughness)?

- a) Mean deviation in height value (or root mean squared)
- b) Maximum peak height
- c) Average distance between highest peak and lowest valley
- d) Roughness factor (used in the Wenzel wetting model)

*** Write the answer in the Answer Book**

Question 1. Answer ALL Parts of Question**Part (a)**

- i. Surface and interface are important phenomena in understanding the interaction of materials. Define surface and interface and explain how they differ geometrically. Also, define the interfacial layer and describe how it affects the properties of materials at both the macro and nano levels. **[6 marks]**
- ii. The adsorption of molecules onto a surface is essential for any surface-mediated chemical process. Explain the five steps in molecular adsorption with an example of a surface-catalyzed reaction of CH_4 in the formation of Graphene. Include a diagram, labelling each stage from the initial to the final stage of the process. **[4 marks]**
- iii. Define a surfactant and explain the terms hydrophilic and hydrophobic. Describe how surfactants arrange themselves in a soap bubble and explain how this arrangement contributes to its stability. **[7 marks]**

Part (b)

- i. Hydrogen bonding plays a critical role in stabilising molecular structures such as water. Explain how hydrogen bonds form and describe their significance in the structure and stability of water molecules. How does hydrogen bonding relate to organic chemistry? **[4 marks]**
- ii. The critical micelle concentration (CMC) is essential for the formation of micelles by amphiphilic molecules. Define CMC and explain its role in micelle formation. How is CMC reached, and how is it monitored in water? Include a diagram of surface tension vs concentration to illustrate your explanation. **[4 marks]**
- iii. Adsorption is the process by which molecules bind to the surface of a solid or liquid. Define adsorption and differentiate it from absorption. Explain the role of the adsorbent and adsorbate in this process. **[4 marks]**
- iv. The nature of the adsorbate and adsorbent interaction determines whether physisorption or chemisorption occurs. Compare physisorption and chemisorption in terms of the binding forces involved and energy requirements. **[4 marks]**

Question 2. Answer ALL Parts of Question**Part (a)**

Physical Vapour Deposition (PVD) can be activated by heat (Thermal PVD) or by an electron beam (e-beam PVD).

- i. Briefly describe why this process has the term “Physical” in its name. Include in your answer a description of the changes occurring in a typical deposition process. **[4 marks]**
- ii. These types of PVD are usually conducted using metals within a vacuum chamber. Explain why this choice of materials and reaction vessel are typically chosen. **[4 marks]**
- iii. Sputtering, another type of PVD, does not require heat for the deposition. Name what is used instead and explain how this process generates a surface coating. **[6 marks]**
- iv. For any of the PVD methods discussed in this question explain how the deposition rate and film thickness is typically controlled. **[3 marks]**

Part (b)

Surface wetting can be predicted/explained using fundamental reasoning and implementing surface wetting models.

- i. For a flat surface, name the factor that principally determines that the water contact angle (WCA) is achieved. **[1 mark]**
- ii. Explain how the factor above (Q2bi) can result in a high ($> 90^\circ$) and a low ($< 90^\circ$) WCA. Make sure to justify your answer. **[4 marks]**
- iii. For the flat surface with either higher or low WCAs above (Q2bii). Provide the name of the classification given to its wetting, in addition, briefly indicate what happens to the WCAs of the surface when it is roughened. **[2 marks]**
- iv. Using the Wenzel wetting model, calculate the WCA achieved on a surface with a Young's WCA of 95° and a roughness factor of 2. Ensure you include all calculation steps. **[4 marks]**
- v. Using the surface material above (Q2biv – Young's WCA = 95°), demonstrate *via* calculation how this can be made superhydrophobic. You are able to use any possible surface wetting model. **[5 marks]**

Question 3. Answer EITHER Part (a) or Part (b) for this Question**Part (a)**

- i. Describe the Langmuir-Blodgett (LB) film technique and its role in forming two-dimensional (2D) layers on surfaces. Explain the process of transferring these layers to substrates and how self-assembly is driven. In your answer, detail the steps involved in film formation and how amphiphilic molecules organize during this process. **[6 marks]**
- ii. Explain the DLVO theory in the context of colloidal stability. What do the terms V_m (barrier) and V_{minimum} represent, and how do they relate to the interaction between two interfaces? Discuss how interfacial charge and the addition of electrolytes affect these values. Include a diagram to illustrate the attractive and repulsive forces involved. **[8 marks]**

Part (b)

A comprehensive material analysis protocol includes the analysis of composition, surface morphology, and localised surface chemistry.

- i. In the characterisation of thin coatings (e.g. Self-assembled monolayers), describe the factors required to ensure the accuracy (validity) of any analysis. **[3 marks]**
- ii. Energy Dispersive X-Ray (EDX) spectroscopy is not considered to be suited to localised surface analysis. Briefly explain why this is the case. **[3 marks]**
- iii. Describe how EDX is able to analyse the composition of a material. Your answer must identify the probing particle or radiation, the nature of the interaction with the material, and what is analysed as a result. **[4 marks]**
- iv. The three-dimensional analysis of surface morphology can be challenging particularly on the micro and nanoscale materials. Name and briefly describe a technique for analysing morphology on this scale. Your answer must include a description of how the measurement is able to quantify the scale of surface features. **[4 marks]**

End of exam paper